

JEPA-Based World Model for Secrecy-Aware Power Control in HAP–UAV Wireless Networks

Complete System, Dataset, Architecture, and Simulation Specification

Research Design Draft

August 2026

Abstract

This document specifies a complete simulation-ready research framework that combines three ideas: (i) the Vision–Memory–Controller decomposition of the World Models framework, (ii) a WirelessJEPA-style self-supervised encoder operating on multi-antenna raw I/Q samples, and (iii) a simplified secrecy–freshness wireless environment derived from an RIS-assisted HAP–IoT model by removing the RIS and retaining only source transmit-power and HAP jamming-power control. The proposed agent observes a four-antenna pilot I/Q window at a legitimate HAP receiver, compresses the radio-frequency observation using JEPA, combines it with a small set of low-dimensional network-state variables, models temporal dynamics using an action-conditioned MDN-LSTM, and allocates source transmit power and HAP jamming power through a compact controller. Age of Information (AoI), Age of Leaked Information (AoLI), and secrecy rate are the principal performance metrics. A separate exhaustive-search oracle is used to generate expert power-allocation labels for behavior-cloning initialization, while sequential trajectories are used to train the world model. The document explicitly distinguishes choices inherited from the source papers from new assumptions introduced to make the simplified model internally consistent and directly implementable.

Contents

1	Research Objective and Design Philosophy	4
1.1	Core hypothesis	4
1.2	What is inherited and what is newly proposed	4
2	System Overview	4
2.1	Network entities	4
2.2	Discrete-time operation	5
3	Geometry and Mobility	5
3.1	Coordinate system	5
3.2	UAV mobility	6
4	Wireless Channel Model	6
4.1	Large-scale path loss	6
4.2	Small-scale fading	7
4.3	Four-element HAP receive array	7
4.4	Residual self-interference and jamming links	7

5	Noise Power	7
6	Pilot I/Q Observation Model	8
6.1	Known pilot sequence	8
6.2	I/Q tensor	8
7	Data-Phase Signal, SINR, Rate, and Secrecy	9
7.1	Control action	9
7.2	Legitimate HAP SINR	9
7.3	UAV eavesdropper SINR	9
7.4	Achievable rates	9
8	Packet Reception, AoI, and AoLI	10
8.1	Fresh update generation	10
8.2	Secrecy gate	10
8.3	Reception indicators	10
8.4	AoI dynamics	10
8.5	AoLI dynamics	10
9	Optimization Problem	11
9.1	Long-term objective	11
10	Environment State, Agent Observation, and Reward	11
10.1	Privileged simulator state	11
10.2	Agent-observable information	12
10.3	Reward	12
11	Dataset Construction	12
11.1	Dataset A: self-supervised JEPA pretraining	12
11.2	Dataset B: world-model trajectories	13
11.3	Dataset C: exhaustive-search expert trajectories	13
12	WirelessJEPA Representation Model	14
12.1	Input reshaping	14
12.2	Context encoder, predictor, and EMA target encoder	14
12.3	Mask geometries	14
12.4	JEPA loss	15
12.5	Inference-time RF latent vector	15
13	Side-Information Encoder and Joint Latent State	15
14	Action-Conditioned World Model	16
14.1	MDN-LSTM memory model	16
14.2	Mixture-density output	16
14.3	Why an MDN remains useful in this wireless environment	17
14.4	Optional auxiliary heads	17

15 Controller	17
15.1 World-Models-faithful linear controller	17
15.2 Behavior-cloning initialization	17
15.3 Optional policy fine-tuning	18
16 Full Proposed Architecture	18
17 Training Pipeline	18
17.1 Stage 0: simulator validation	18
17.2 Stage 1: collect broad exploration trajectories	19
17.3 Stage 2: JEPA pretraining	19
17.4 Stage 3: encode trajectories and train MDN-LSTM	19
17.5 Stage 4: generate exhaustive-search expert trajectories	19
17.6 Stage 5: behavior-clone the controller	19
17.7 Stage 6: optional long-horizon fine-tuning	19
17.8 Stage 7: optional dream-environment training	19
18 Algorithms	20
19 Reference Dimensions	20
20 Reference Simulation Parameters	20
21 VAE Baseline for the Central Comparison	20
22 Evaluation Baselines and Ablations	20
22.1 Required baselines	20
22.2 JEPA-specific ablations	21
22.3 World-model ablations	22
23 Performance Metrics	23
23.1 Wireless control metrics	23
23.2 Representation metrics	23
23.3 World-model metrics	23
23.4 Control-learning metrics	24
24 Recommended Experimental Progression	24
25 Implementation Sanity Checks	24
25.1 Unit checks	25
25.2 Causality checks	25
25.3 No privileged-state leakage	25
25.4 Data leakage checks	25
26 Open Design Decisions to Revisit After the First Smoke Runs	25
27 Minimum Reproducible Experiment	26
28 Summary of the Proposed Scientific Contribution	26

1 Research Objective and Design Philosophy

1.1 Core hypothesis

The original World Models architecture divides the agent into three components: a representation model V , a memory/dynamics model M , and a controller C [1]. The V block compresses the high-dimensional observation into a compact latent representation; the M block predicts the future latent state conditioned on the current latent state, action, and recurrent memory; and the C block maps the latent representation and recurrent hidden state to the action.

The central hypothesis of the present work is

$$\boxed{\text{WirelessJEPA representation} \Rightarrow \text{more control-relevant/predictable wireless latent state} \Rightarrow \text{better world-model-}} \quad (1)$$

relative to a reconstruction-based VAE representation under an otherwise matched world-model and controller design.

Accordingly, the proposed architecture replaces the VAE representation model by a JEPA encoder while retaining the World Models decomposition:

$$\text{I/Q observation} \xrightarrow{\text{JEPA}} \mathbf{z}_t \xrightarrow[\mathbf{a}_t]{\text{MDN-LSTM}} \mathbf{h}_{t+1} \xrightarrow{\text{Controller}} \mathbf{a}_{t+1}. \quad (2)$$

1.2 What is inherited and what is newly proposed

To avoid ambiguity, the following distinction is maintained throughout the document.

- **Inherited from World Models [1]:** separate representation, recurrent world model, and compact controller; action-conditioned prediction of the next latent vector; MDN-LSTM for stochastic latent dynamics; controller input formed from latent state and recurrent hidden state; separate training of representation and dynamics models as the default implementation strategy.
- **Inherited from WirelessJEPA [2]:** raw multi-antenna I/Q input, $2 \times 4 \times 256$ sample geometry, antenna-dimension upsampling to a $2 \times 256 \times 256$ grid, structured antenna/time/multi-block masking, a context encoder, a lightweight predictor, an exponential-moving-average target encoder, and an L2 latent-prediction loss on masked regions.
- **Inherited from the intent-driven HAP-IoT model [3]:** HAP legitimate receiver, UAV eavesdropper, full-duplex HAP artificial jamming, mmWave path-loss structure, Nakagami fading, residual self-interference, AoI/AoLI objectives, secrecy-rate calculation, and transmit/jamming power adaptation.
- **New design choices in this document:** no RIS; four receive antennas at the HAP for I/Q observation; a pilot-before-decision slot structure to remove causality ambiguity; explicit multi-antenna pilot I/Q generation; a compact scalar side-information vector; a JEPA projection dimension; a joint JEPA/side-information latent state; exhaustive-search behavior-cloning labels; proposed reward normalization; and a staged training/evaluation protocol.

2 System Overview

2.1 Network entities

The simplified wireless system contains three physical nodes:

- (i) a ground source/user S generating status-update packets;
- (ii) a HAP H serving as the legitimate receiver and artificial-noise jammer;
- (iii) a mobile/hovering UAV U acting as a passive eavesdropper.

The RIS in [3] is removed entirely. Therefore, the only control variables are source transmit power and HAP jamming power.

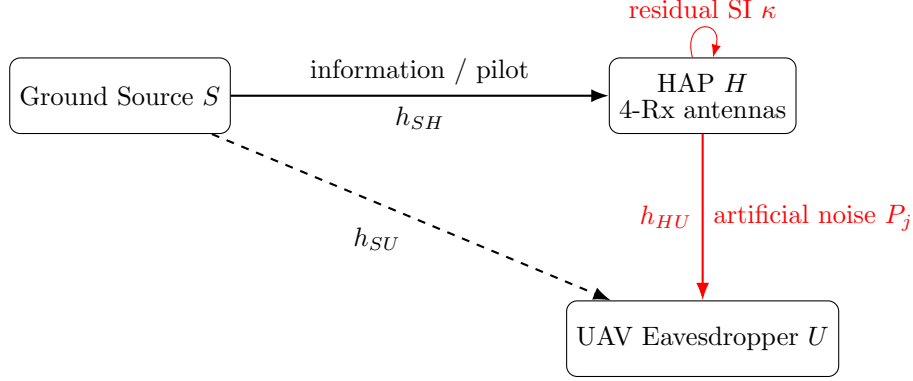


Figure 1: Proposed simplified HAP–UAV secrecy environment. The HAP receives the source update and simultaneously jams the eavesdropper during the data phase.

2.2 Discrete-time operation

Time is divided into slots $t \in \{0, 1, \dots, T-1\}$ of duration T_s . Each slot is split into two logical phases:

$$T_s = T_{\text{pilot}} + T_{\text{data}}. \quad (3)$$

The key reason for this split is causality. The controller must not use I/Q samples that are themselves generated by the same power action it has not yet selected. Therefore:

1. **Pilot/observation phase:** the source transmits a known pilot at fixed pilot power P_p ; the HAP jamming transmitter is silent; the four HAP receive antennas collect a raw I/Q window $\mathbf{x}_t^{\text{IQ}}$.
2. **Decision phase:** JEPA encodes the pilot I/Q window; the controller selects $\mathbf{a}_t = [P_t(t), P_j(t)]^\top$.
3. **Data/jamming phase:** the source transmits the fresh update using $P_t(t)$ while the HAP emits artificial noise using $P_j(t)$; the environment computes legitimate rate, eavesdropper rate, secrecy rate, packet outcomes, AoI, AoLI, reward, and the next state.

3 Geometry and Mobility

3.1 Coordinate system

Let the source be fixed at

$$\mathbf{p}_S = [0, 0, 0]^\top. \quad (4)$$

The HAP is fixed at

$$\mathbf{p}_H = [x_H, y_H, H_H]^\top, \quad (5)$$

with default $x_H = y_H = 0$ and altitude $H_H = 20$ km, consistent with the scale used in [3]. The UAV position is

$$\mathbf{p}_U(t) = [x_U(t), y_U(t), H_U]^\top. \quad (6)$$

The Euclidean distance between nodes i and j is

$$d_{ij}(t) = \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\|_2. \quad (7)$$

Distances are converted to km before being used in the path-loss expression below.

3.2 UAV mobility

For the initial static experiment,

$$\mathbf{p}_U(t) = \mathbf{p}_U(0), \quad \forall t. \quad (8)$$

A dynamic extension uses constant-velocity or random-walk motion:

$$\mathbf{p}_U(t+1) = \mathbf{p}_U(t) + T_s \mathbf{v}_U(t), \quad (9)$$

where $\mathbf{v}_U(t) = [v_x(t), v_y(t), 0]^\top$. For a random-walk trajectory, the heading is periodically resampled; for a circular trajectory, $\mathbf{p}_U(t)$ is parameterized by angular position and angular velocity. These two mobility families are motivated by the robustness tests used in [3].

4 Wireless Channel Model

4.1 Large-scale path loss

For a link $i \rightarrow j$ separated by d_{ij} km at carrier frequency f_c GHz, use the path-loss model adopted in [3]:

$$L_{ij}(t) [\text{dB}] = 92.45 + 20 \log_{10}(f_c) + 20 \log_{10}(d_{ij}(t)) + \alpha_{\text{atm}} d_{ij}(t) + \zeta_{ij}(t), \quad (10)$$

where α_{atm} is atmospheric attenuation in dB/km and ζ_{ij} is a shadowing/extra-loss term. The corresponding linear large-scale gain is

$$\beta_{ij}(t) = 10^{-L_{ij}(t)/10}. \quad (11)$$

Antenna and hardware gains can be included through

$$\bar{\beta}_{ij}(t) = \beta_{ij}(t) G_i G_j \ell_i \ell_j, \quad (12)$$

where G_i, G_j are linear antenna gains and $\ell_i, \ell_j \in (0, 1]$ represent hardware-loss factors. If gains/losses are already included in a link budget, they must not be applied a second time.

4.2 Small-scale fading

Following [3], the baseline simulation uses Nakagami- m fading. For scalar link $i \rightarrow j$,

$$h_{ij}(t) = \sqrt{\bar{\beta}_{ij}(t)} \tilde{h}_{ij}(t), \quad (13)$$

where the envelope $|\tilde{h}_{ij}|$ is Nakagami distributed with shape parameter m_{ij} and normalized second moment $\mathbb{E}[|\tilde{h}_{ij}|^2] = 1$. A uniformly distributed phase can be added:

$$\tilde{h}_{ij}(t) = r_{ij}(t)e^{j\varphi_{ij}(t)}, \quad \varphi_{ij} \sim \mathcal{U}[0, 2\pi). \quad (14)$$

The initial simulator may use block fading, i.e., channels are constant over one slot. To provide meaningful temporal structure for the recurrent world model, later experiments should include slowly varying geometry and/or temporally correlated fading rather than fully independent geometry and fading in every slot.

4.3 Four-element HAP receive array

The HAP employs $M_H = 4$ receive antennas in a uniform linear array (ULA) with inter-element spacing $d_a = \lambda/2$, where $\lambda = c/f_c$. Let $\vartheta_{SH}(t)$ denote the source angle of arrival. The normalized steering vector is

$$\mathbf{a}_H(\vartheta) = \frac{1}{\sqrt{M_H}} \begin{bmatrix} 1 & e^{-j2\pi \frac{d_a}{\lambda} \sin \vartheta} & \dots & e^{-j2\pi(M_H-1) \frac{d_a}{\lambda} \sin \vartheta} \end{bmatrix}^T. \quad (15)$$

The source-to-HAP channel vector is

$$\mathbf{h}_{SH}(t) = \sqrt{\bar{\beta}_{SH}(t)} \tilde{h}_{SH}(t) \mathbf{a}_H(\vartheta_{SH}(t)). \quad (16)$$

This model creates physically meaningful cross-antenna phase structure for JEPA while keeping the overall propagation model simple. More detailed multipath can later be added as a sum of L_p paths.

4.4 Residual self-interference and jamming links

The HAP is full duplex during the data phase. Let h_{HH} denote the effective self-interference channel before cancellation and let $\kappa \in [0, 1]$ be the residual self-interference factor after cancellation. The effective residual interference power is proportional to $\kappa P_j |h_{HH}|^2$.

The HAP-to-UAV jamming channel is $h_{HU}(t)$. If directional jamming is later modeled, $|h_{HU}|^2$ can be replaced by the corresponding beamformed gain.

5 Noise Power

Let the one-sided thermal noise density be $N_0 = -174$ dBm/Hz and receiver noise figure be NF dB. The total receiver noise power over bandwidth B is

$$\sigma_{\text{dBm}}^2 = N_0 + 10 \log_{10}(B) + NF, \quad (17)$$

with B in Hz. The corresponding linear power in watts is

$$\sigma^2 = 10^{(\sigma_{\text{dBm}}^2 - 30)/10}. \quad (18)$$

6 Pilot I/Q Observation Model

6.1 Known pilot sequence

At the beginning of every slot, the source transmits a known length- W complex pilot sequence

$$\mathbf{s}_p = [s_p[0], \dots, s_p[W-1]], \quad \frac{1}{W} \sum_{n=0}^{W-1} |s_p[n]|^2 = 1, \quad (19)$$

using a fixed power P_p . The HAP jammer is silent in the pilot phase.

For antenna $m \in \{0, \dots, M_H - 1\}$ and sample n , the received pilot sample is

$$r_{m,t}[n] = \sqrt{P_p} h_{SH,m}(t) s_p[n - \tau_t] e^{j(2\pi \Delta f_t n T_{\text{samp}} + \phi_t)} + w_{m,t}[n], \quad (20)$$

where

- τ_t is an optional integer timing offset;
- Δf_t is an optional residual carrier-frequency offset (CFO);
- ϕ_t is a random carrier phase;
- $w_{m,t}[n] \sim \mathcal{CN}(0, \sigma_p^2)$.

The baseline can start with $\tau_t = 0$ and $\Delta f_t = 0$ and enable them later as domain-randomization variables. WirelessJEPA intentionally retains realistic timing/CFO imperfections in its over-the-air dataset [2]; therefore, adding such impairments is a reasonable robustness extension but is not mandatory for the first simulator.

6.2 I/Q tensor

The raw complex matrix is

$$\mathbf{R}_t \in \mathbb{C}^{M_H \times W}. \quad (21)$$

Convert it into the real-valued tensor

$$\mathbf{x}_t^{\text{IQ}} \in \mathbb{R}^{2 \times M_H \times W}, \quad (22)$$

where

$$\mathbf{x}_t^{\text{IQ}}[0, m, n] = \Re\{r_{m,t}[n]\}, \quad (23)$$

$$\mathbf{x}_t^{\text{IQ}}[1, m, n] = \Im\{r_{m,t}[n]\}. \quad (24)$$

For $M_H = 4$ and $W = 256$,

$$\boxed{\mathbf{x}_t^{\text{IQ}} \in \mathbb{R}^{2 \times 4 \times 256}}. \quad (25)$$

Following WirelessJEPA, unit-max normalization is applied per sample:

$$\mathbf{x}_t^{\text{IQ}} \leftarrow \frac{\mathbf{x}_t^{\text{IQ}}}{\max |\mathbf{x}_t^{\text{IQ}}| + \epsilon}, \quad (26)$$

where ϵ prevents division by zero.

7 Data-Phase Signal, SINR, Rate, and Secrecy

7.1 Control action

At slot t , the controller selects

$$\mathbf{a}_t = \begin{bmatrix} P_t(t) \\ P_j(t) \end{bmatrix}, \quad (27)$$

with independent physical power budgets

$$0 \leq P_t(t) \leq P_t^{\max}, \quad (28)$$

$$0 \leq P_j(t) \leq P_j^{\max}. \quad (29)$$

Independent budgets are used because the source transmitter and HAP jammer are physically distinct nodes. This differs from the shared-budget expression in [3] and is a deliberate simplification/clarification for the proposed system.

7.2 Legitimate HAP SINR

During the data phase, maximum-ratio combining (MRC) can be used at the HAP. Let

$$\mathbf{w}_H(t) = \frac{\mathbf{h}_{SH}(t)}{\|\mathbf{h}_{SH}(t)\|_2}. \quad (30)$$

With equivalent residual self-interference gain $G_{HH}(t)$, the post-combining HAP SINR is

$$\gamma_H(t) = \frac{P_t(t)\|\mathbf{h}_{SH}(t)\|_2^2}{\sigma_H^2 + \kappa P_j(t)G_{HH}(t)}. \quad (31)$$

If the receive array gain is already included in G_H in the link budget, use a normalized array response to avoid double counting.

7.3 UAV eavesdropper SINR

For a single-antenna UAV,

$$\gamma_U(t) = \frac{P_t(t)|h_{SU}(t)|^2}{\sigma_U^2 + P_j(t)|h_{HU}(t)|^2}. \quad (32)$$

7.4 Achievable rates

The instantaneous rates are

$$R_H(t) = B \log_2(1 + \gamma_H(t)), \quad (33)$$

$$R_U(t) = B \log_2(1 + \gamma_U(t)). \quad (34)$$

The instantaneous secrecy rate is

$$R_s(t) = [R_H(t) - R_U(t)]^+, \quad (35)$$

where $[x]^+ = \max(x, 0)$.

8 Packet Reception, AoI, and AoLI

8.1 Fresh update generation

The source follows a generate-at-will model: one fresh packet of size L_{pkt} bits is available at the start of each slot. Define the required data-phase rate

$$R_{\text{req}} = \frac{L_{\text{pkt}}}{T_{\text{data}}}. \quad (36)$$

8.2 Secrecy gate

To remain close to the secrecy-gated mechanism in [3], define

$$g_t = \mathbb{I}\{R_s(t) \geq R_s^{\min}\}. \quad (37)$$

If $g_t = 0$, the confidential data update is withheld and a dummy/idle transmission is assumed. This produces a clear security–freshness tradeoff: overly conservative power allocation protects secrecy but increases AoI.

8.3 Reception indicators

The legitimate-reception indicator is

$$\mu_H(t) = g_t \mathbb{I}\{R_H(t) \geq R_{\text{req}}\}. \quad (38)$$

For AoLI tracking, define an eavesdropper decoding indicator

$$\mu_U(t) = g_t \mathbb{I}\{R_U(t) \geq R_U^{\text{dec}}\}. \quad (39)$$

Here R_U^{dec} is the decoding threshold used for the AoLI abstraction. This is a simulation-level definition: R_s remains the physical-layer secrecy metric, while μ_U determines whether the eavesdropper receives a sufficiently informative packet to reset AoLI.

8.4 AoI dynamics

Let $\Delta_H(t)$ denote the age of the newest successfully received legitimate update at the beginning of slot t . For generate-at-will traffic with one-slot normalization,

$$\Delta_H(t+1) = \begin{cases} 1, & \mu_H(t) = 1, \\ \min\{\Delta_H(t) + 1, \Delta_{\max}\}, & \mu_H(t) = 0. \end{cases} \quad (40)$$

This is the simplified one-slot version of the generation-time update used in [3].

8.5 AoLI dynamics

Let $\Delta_U(t)$ denote the age of the freshest information available to the UAV. Larger AoLI is desirable. Use

$$\Delta_U(t+1) = \begin{cases} 1, & \mu_U(t) = 1, \\ \min\{\Delta_U(t) + 1, \Delta_{\max}\}, & \mu_U(t) = 0. \end{cases} \quad (41)$$

9 Optimization Problem

9.1 Long-term objective

Over horizon T , define

$$\bar{\Delta}_H = \frac{1}{T} \sum_{t=0}^{T-1} \Delta_H(t), \quad (42)$$

$$\bar{\Delta}_U = \frac{1}{T} \sum_{t=0}^{T-1} \Delta_U(t), \quad (43)$$

$$\bar{R}_s = \frac{1}{T} \sum_{t=0}^{T-1} R_s(t), \quad (44)$$

$$\bar{P}_t = \frac{1}{T} \sum_{t=0}^{T-1} P_t(t), \quad (45)$$

$$\bar{P}_j = \frac{1}{T} \sum_{t=0}^{T-1} P_j(t). \quad (46)$$

A primary optimization objective consistent with the freshness–secrecy direction of [3] is

$$\min_{\pi} \quad \bar{\Delta}_H - \omega \bar{\Delta}_U + \lambda_t \bar{P}_t + \lambda_j \bar{P}_j, \quad (47)$$

subject to

$$0 \leq P_t(t) \leq P_t^{\max}, \quad (48)$$

$$0 \leq P_j(t) \leq P_j^{\max}, \quad (49)$$

$$\mathbf{a}_t = \pi(\mathcal{O}_t), \quad (50)$$

where $\mathcal{O}_t$ is the agent observation/history. Secrecy is enforced operationally by g_t and can additionally be reported as a violation/outage metric.

An alternative constrained formulation is

$$\min_{\pi} \quad \bar{\Delta}_H - \omega \bar{\Delta}_U + \lambda_t \bar{P}_t + \lambda_j \bar{P}_j \quad (51)$$

$$\text{s.t.} \quad \Pr\{R_s(t) < R_s^{\min}\} \leq \epsilon_s, \quad (52)$$

which is useful for later constrained-RL experiments.

10 Environment State, Agent Observation, and Reward

10.1 Privileged simulator state

The simulator has access to a full state

$$\mathbf{s}_t^{\text{env}} = \{\mathbf{p}_U(t), \mathbf{v}_U(t), \mathbf{h}_{SH}(t), h_{SU}(t), h_{HU}(t), h_{HH}(t), \Delta_H(t), \Delta_U(t)\}. \quad (53)$$

This privileged state is used for physics, exhaustive search, and evaluation; it is not necessarily exposed to the controller.

10.2 Agent-observable information

The high-dimensional RF observation is the HAP pilot tensor $\mathbf{x}_t^{\text{IQ}}$. Because the HAP pilot observation primarily describes the $S \rightarrow H$ link, it cannot by itself uniquely reveal the passive eavesdropper links $S \rightarrow U$ and $H \rightarrow U$. Therefore, the proposed controller also receives a small side-information vector representing information that is assumed available from tracking/sensing or a network-state estimator:

$$\mathbf{q}_t = [\tilde{\Delta}_H(t), \tilde{\Delta}_U(t), \tilde{\beta}_{SU}(t), \tilde{\beta}_{HU}(t), \tilde{v}_x(t), \tilde{v}_y(t), \tilde{P}_t(t-1), \tilde{P}_j(t-1)]^\top \in \mathbb{R}^8. \quad (54)$$

Tildes denote normalized values. If UAV velocity is unavailable or the UAV is static, those two entries can be removed. If exact AoLI is considered unrealistic, $\tilde{\Delta}_U$ can later be replaced by an estimated/belief-state AoLI; the first implementation assumes it is observable to match the state abstraction in [3].

10.3 Reward

A normalized reward avoids scale domination by age, rate, or power. Define

$$\tilde{R}_s(t) = \min \left\{ \frac{R_s(t)}{R_s^{\text{ref}}}, 1 \right\}, \quad \tilde{P}_t(t) = \frac{P_t(t)}{P_t^{\text{max}}}, \quad \tilde{P}_j(t) = \frac{P_j(t)}{P_j^{\text{max}}}. \quad (55)$$

A practical reward is

$$\begin{aligned} r_t = & -w_H \frac{\Delta_H(t+1)}{\Delta_{\text{max}}} + w_U \frac{\Delta_U(t+1)}{\Delta_{\text{max}}} + w_s \tilde{R}_s(t) \\ & -w_P \left(\tilde{P}_t(t) + \tilde{P}_j(t) \right) - w_v \mathbb{I}\{R_s(t) < R_s^{\text{min}}\}. \end{aligned} \quad (56)$$

A safe initial set is

$$(w_H, w_U, w_s, w_P, w_v) = (1.0, 0.2, 0.5, 0.05, 1.0), \quad (57)$$

followed by sensitivity analysis. The smaller initial w_U reduces the trivial policy of deliberately withholding all updates solely to increase AoLI.

11 Dataset Construction

Three datasets are required because JEPA pretraining, temporal world-model learning, and expert controller learning solve different learning problems.

11.1 Dataset A: self-supervised JEPA pretraining

The JEPA dataset contains unlabeled I/Q windows:

$$\mathcal{D}_{\text{JEPA}} = \{\mathbf{x}_n^{\text{IQ}}\}_{n=1}^{N_{\text{JEPA}}}. \quad (58)$$

For every sample, randomize a broad set of nuisance and channel variables, e.g.

$$\{\vartheta_{SH}, d_{SH}, m_{SH}, \zeta_{SH}, \phi, \Delta f, \tau, \text{SNR}\}. \quad (59)$$

No optimal-action labels are required. The objective is representation learning from the structure of the I/Q signal.

Important split rule. Train/validation/test splits must be created by trajectories, geometry seeds, or channel-realization groups rather than by randomly shuffling adjacent windows. This prevents near-duplicate temporal samples from leaking across splits.

11.2 Dataset B: world-model trajectories

The recurrent world model needs ordered transitions:

$$\mathcal{D}_{\text{WM}} = \left\{ \left(\mathbf{x}_t^{\text{IQ}}, \mathbf{q}_t, \mathbf{a}_t, r_t, \mathbf{x}_{t+1}^{\text{IQ}}, \mathbf{q}_{t+1} \right) \right\}. \quad (60)$$

The collection policy should deliberately cover the action space. A recommended mixture is

$$\pi_{\text{collect}} = \rho_1 \pi_{\text{random}} + \rho_2 \pi_{\text{grid}} + \rho_3 \pi_{\text{heuristic}} + \rho_4 \pi_{\text{expert}}, \quad (61)$$

with $\sum_i \rho_i = 1$. Initially use mostly random/grid actions to avoid a narrow dataset; after an expert is available, append expert trajectories.

11.3 Dataset C: exhaustive-search expert trajectories

For each environment state encountered in an expert rollout, evaluate all pairs on a discrete power grid

$$\mathcal{P}_t = \{0, \Delta P_t, 2\Delta P_t, \dots, P_t^{\max}\}, \quad (62)$$

$$\mathcal{P}_j = \{0, \Delta P_j, 2\Delta P_j, \dots, P_j^{\max}\}. \quad (63)$$

The candidate set is

$$\mathcal{A}_{\text{grid}} = \mathcal{P}_t \times \mathcal{P}_j. \quad (64)$$

With 0–30 dBm and 1 dB spacing, $|\mathcal{A}_{\text{grid}}| = 31^2 = 961$ actions per state.

For each candidate, clone the current simulator state, execute one counterfactual step, and evaluate

$$J_t(\mathbf{a}) = r_t(\mathbf{a}) \quad (65)$$

for the one-step oracle. The expert label is

$$\mathbf{a}_t^* = \arg \max_{\mathbf{a} \in \mathcal{A}_{\text{grid}}} J_t(\mathbf{a}). \quad (66)$$

The selected expert action is then applied to the non-cloned environment so that the next expert sample remains on a coherent expert trajectory.

The resulting dataset is

$$\mathcal{D}_{\text{ES}} = \left\{ \mathbf{x}_t^{\text{IQ}}, \mathbf{q}_t, \mathbf{a}_t^*, r_t^*, \mathbf{x}_{t+1}^{\text{IQ}}, \mathbf{q}_{t+1} \right\}_{t=0}^{T-1}. \quad (67)$$

Limitation of one-step exhaustive search. The label in (66) is myopic. AoI and AoLI are inherently sequential, so one-step exhaustive search is an expert initialization and benchmark rather than a guaranteed optimum of the finite-horizon control problem. A later short-horizon oracle may use

$$\mathbf{a}_t^* = \arg \max_{\mathbf{a}_t} \max_{\mathbf{a}_{t+1:t+H-1}} \sum_{\ell=0}^{H-1} \gamma^\ell r_{t+\ell}, \quad (68)$$

but naive complexity becomes $|\mathcal{A}|^H$; beam search or model-predictive control is more practical for $H > 1$.

12 WirelessJEPA Representation Model

12.1 Input reshaping

WirelessJEPA uses multi-antenna raw I/Q samples with shape $2 \times 4 \times 256$ and upsamples the antenna dimension by nearest-neighbor repetition to obtain a square antenna–time plane [2]. We retain the same preprocessing for the first controlled experiment:

$$\mathbf{x}_{\text{raw}} \in \mathbb{R}^{2 \times 4 \times 256} \longrightarrow \mathbf{x} \in \mathbb{R}^{2 \times 256 \times 256}. \quad (69)$$

Specifically,

$$x_{c,i,n} = (x_{\text{raw}})_{c,\lfloor i/64 \rfloor, n}, \quad i, n \in \{0, \dots, 255\}. \quad (70)$$

The upsampling is retained to reproduce the WirelessJEPA design as closely as possible. It should later be ablated against a native $2 \times 4 \times 256$ architecture because repetition adds no new physical information.

12.2 Context encoder, predictor, and EMA target encoder

Define

- context encoder E_θ ;
- predictor P_ϕ ;
- target/teacher encoder $E_{\tilde{\theta}}$.

Following WirelessJEPA, the context encoder uses ShuffleNetV2-0.5 $\times$, the target encoder shares the same architecture but operates densely, and the predictor uses lightweight depthwise-separable convolutions [2].

For a latent-space mask $\mathbf{M}_{\text{lat}}$, the context input is masked and sparse activations are maintained by reapplying an appropriately resized mask after convolution/pooling layers. The context feature map is

$$\mathbf{H}_t = E_\theta(\mathbf{x}_t, \mathbf{M}_{\text{lat}}). \quad (71)$$

A learnable mask token is inserted at masked latent locations to form $\tilde{\mathbf{H}}_t$, and the predictor produces

$$\hat{\mathbf{Y}}_t = P_\phi(\tilde{\mathbf{H}}_t). \quad (72)$$

The target encoder receives the full unmasked input:

$$\mathbf{Y}_t = E_{\tilde{\theta}}(\mathbf{x}_t). \quad (73)$$

12.3 Mask geometries

The three structured masks of interest are:

1. **Antenna mask:** hides one or more complete antenna bands; encourages cross-antenna/spatial inference.
2. **Time mask:** hides contiguous temporal spans; encourages temporal/waveform inference.
3. **Multi-block mask:** hides multiple contiguous antenna–time blocks; encourages mixed spatial–temporal inference.

WirelessJEPA reports that mask geometry produces materially different inductive biases [2]. For the proposed control application, multi-block masking is the default because both temporal signal structure and cross-antenna phase structure may matter; antenna and time masking are required ablations.

A source-aligned initial patch configuration is

$$\text{multi-block: } 64 \times 32, \quad (74)$$

$$\text{antenna: } 64 \times 256, \quad (75)$$

$$\text{time: } 256 \times 32. \quad (76)$$

12.4 JEPA loss

Let $\mathcal{M}$ be the set of masked latent indices. The pretraining loss is

$$\mathcal{L}_{\text{JEPA}} = \frac{1}{|\mathcal{M}|} \sum_{(i,j) \in \mathcal{M}} \left\| \hat{\mathbf{Y}}_{t,i,j} - \mathbf{Y}_{t,i,j} \right\|_2^2. \quad (77)$$

Gradients update only θ and ϕ . The target parameters are updated by EMA:

$$\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau) \theta. \quad (78)$$

WirelessJEPA uses an EMA coefficient scheduled from approximately 0.996 toward 1.0, AdamW, cosine learning-rate decay, and 100 pretraining epochs [2]; those are the recommended starting settings here.

12.5 Inference-time RF latent vector

After pretraining, the predictor is discarded and the context encoder is retained. At inference, use an all-visible mask and obtain a full feature map

$$\mathbf{H}_t^{\text{full}} = E_{\theta}(\mathbf{x}_t, \mathbf{1}). \quad (79)$$

Apply global average pooling followed by a learned projection:

$$\mathbf{z}_t^{\text{RF}} = W_{\text{RF}} \text{GAP}(\mathbf{H}_t^{\text{full}}) + \mathbf{b}_{\text{RF}}, \quad \mathbf{z}_t^{\text{RF}} \in \mathbb{R}^{d_{\text{RF}}}. \quad (80)$$

The WirelessJEPA paper does not prescribe the compact downstream latent dimension required by the present world model. We therefore propose

$$\boxed{d_{\text{RF}} = 128} \quad (81)$$

as a starting value, with 64, 128, and 256 included in an ablation.

13 Side-Information Encoder and Joint Latent State

The scalar side-information vector $\mathbf{q}_t \in \mathbb{R}^8$ is not passed through JEPA. It is encoded by a small MLP:

$$\mathbf{e}_t = \text{ReLU}(W_1 \mathbf{q}_t + \mathbf{b}_1), \quad (82)$$

$$\mathbf{z}_t^{\text{net}} = W_2 \mathbf{e}_t + \mathbf{b}_2, \quad \mathbf{z}_t^{\text{net}} \in \mathbb{R}^{d_{\text{net}}}. \quad (83)$$

Use $d_{\text{net}} = 32$ initially.

The complete latent state supplied to the world model and controller is

$$\boxed{\mathbf{z}_t = \begin{bmatrix} \mathbf{z}_t^{\text{RF}} \\ \mathbf{z}_t^{\text{net}} \end{bmatrix} \in \mathbb{R}^{d_z}, \quad d_z = d_{\text{RF}} + d_{\text{net}} = 160.} \quad (84)$$

This separation is intentional: JEPA handles the high-dimensional radio observation, while a small MLP handles quantities that are already compact and should not be artificially expanded into an image-like representation.

14 Action-Conditioned World Model

14.1 MDN-LSTM memory model

World Models uses an LSTM with a mixture-density output to model the stochastic distribution of the next latent vector [1]. We retain this principle. Let the normalized action be

$$\tilde{\mathbf{a}}_t = \begin{bmatrix} P_t(t)/P_t^{\text{max}} \\ P_j(t)/P_j^{\text{max}} \end{bmatrix} \in [0, 1]^2. \quad (85)$$

The recurrent input is

$$\mathbf{u}_t = [\mathbf{z}_t; \tilde{\mathbf{a}}_t] \in \mathbb{R}^{d_z+2}. \quad (86)$$

The LSTM update is

$$(\mathbf{h}_{t+1}, \mathbf{c}_{t+1}) = \text{LSTM}_\psi(\mathbf{u}_t, \mathbf{h}_t, \mathbf{c}_t), \quad (87)$$

where $\mathbf{h}_t, \mathbf{c}_t \in \mathbb{R}^{d_h}$. Use

$$\boxed{d_h = 256} \quad (88)$$

as the initial hidden size, matching the CarRacing MDN-LSTM hidden dimension reported in [1].

14.2 Mixture-density output

Use $K = 5$ Gaussian mixture components, also matching the reference World Models implementation [1]. For each next-latent dimension $d \in \{1, \dots, d_z\}$, the model outputs

$$\{\pi_{d,k,t}, \mu_{d,k,t}, \sigma_{d,k,t}\}_{k=1}^K, \quad (89)$$

where

$$\pi_{d,k,t} = \text{softmax}_k(\ell_{d,k,t}), \quad (90)$$

$$\sigma_{d,k,t} = \text{softplus}(s_{d,k,t}) + \sigma_{\text{min}}. \quad (91)$$

A factored mixture model is

$$p(\mathbf{z}_{t+1} \mid \mathbf{z}_t, \mathbf{a}_t, \mathbf{h}_t) = \prod_{d=1}^{d_z} \sum_{k=1}^K \pi_{d,k,t} \mathcal{N}(\mathbf{z}_{t+1,d}; \mu_{d,k,t}, \sigma_{d,k,t}^2). \quad (92)$$

The MDN loss is the negative log likelihood

$$\mathcal{L}_{\text{MDN}} = -\frac{1}{N_{\text{seq}}} \sum_{\text{sequences}} \sum_t \log p(\mathbf{z}_{t+1} \mid \mathbf{z}_t, \mathbf{a}_t, \mathbf{h}_t). \quad (93)$$

14.3 Why an MDN remains useful in this wireless environment

Even if geometry evolves deterministically, wireless fading, shadowing, packet outcomes, and imperfectly observed eavesdropper conditions make the latent transition stochastic. A mixture density model can represent multimodal or heteroscedastic uncertainty rather than collapsing to a single deterministic next latent vector.

14.4 Optional auxiliary heads

For the first implementation, the MDN next-latent loss is sufficient. For later imagined rollouts, add explicit heads for reward and termination/security-gate outcomes:

$$\hat{r}_t = f_r(\mathbf{h}_{t+1}), \quad (94)$$

$$\hat{g}_t = \sigma(f_g(\mathbf{h}_{t+1})). \quad (95)$$

An augmented loss is

$$\mathcal{L}_M = \mathcal{L}_{\text{MDN}} + \lambda_r \|\hat{r}_t - r_t\|_2^2 + \lambda_g \text{BCE}(\hat{g}_t, g_t). \quad (96)$$

These heads are not required if controller training is performed only in the real simulator.

15 Controller

15.1 World-Models-faithful linear controller

The controller receives the current latent vector and recurrent hidden state:

$$\mathbf{f}_t = [\mathbf{z}_t; \mathbf{h}_t] \in \mathbb{R}^{d_z+d_h} = \mathbb{R}^{416}. \quad (97)$$

Following the compact-controller principle in [1], use a linear mapping

$$\mathbf{u}_t^C = W_C \mathbf{f}_t + \mathbf{b}_C \in \mathbb{R}^2. \quad (98)$$

Map the two outputs to feasible powers using the sigmoid function:

$$P_t(t) = P_t^{\max} \sigma(u_{t,1}^C), \quad (99)$$

$$P_j(t) = P_j^{\max} \sigma(u_{t,2}^C). \quad (100)$$

If minimum nonzero power levels are needed, use affine scaling between $P^{\min}$ and $P^{\max}$.

15.2 Behavior-cloning initialization

Given exhaustive-search target action $\mathbf{a}_t^*$, normalize the expert powers to $[0, 1]$ and train with

$$\mathcal{L}_{\text{BC}} = \frac{1}{N} \sum_t \|\tilde{\mathbf{a}}_t^C - \tilde{\mathbf{a}}_t^*\|_2^2. \quad (101)$$

During BC training, JEPA and the MDN-LSTM are initially frozen. The recurrent hidden state $\mathbf{h}_t$ is generated by sequentially processing the expert trajectory, not by treating samples as independent.

15.3 Optional policy fine-tuning

One-step exhaustive search is not globally optimal for long-term AoI/AoLI. Therefore, after BC, the controller can be fine-tuned by maximizing

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T-1} \gamma^t r_t \right]. \quad (102)$$

Two compatible choices are:

- **CMA-ES**: closest to the original World Models controller-training method and especially attractive for the small linear controller.
- **PPO**: useful when the controller is enlarged to a small MLP or when later end-to-end policy fine-tuning is desired.

For a clean VAE-versus-JEPA comparison, the controller architecture and controller optimizer must be identical across the two representation models.

16 Full Proposed Architecture

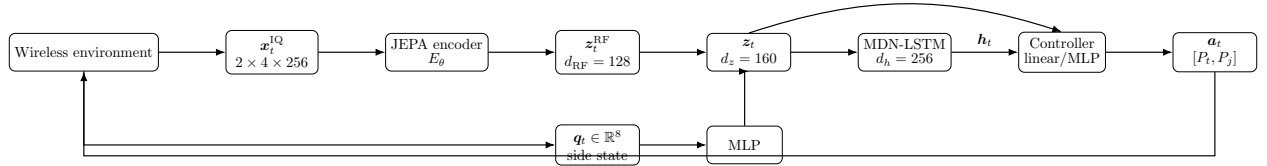


Figure 2: End-to-end deployment architecture. JEPA processes only the high-dimensional pilot I/Q observation; the compact network state is encoded separately. The MDN-LSTM provides memory and stochastic latent dynamics.

17 Training Pipeline

17.1 Stage 0: simulator validation

Before training any neural component, validate the environment numerically:

1. Increasing P_t should generally increase both R_H and R_U .
2. Increasing P_j should generally decrease R_U but, because of residual self-interference, can also reduce R_H .
3. Increasing source-UAV distance should reduce the eavesdropper rate on average.
4. If the HAP update succeeds, AoI should reset; otherwise it should increase by one.
5. If the UAV decodes, AoLI should reset; otherwise it should increase by one.
6. Power and rate units should be checked by hand for several deterministic channel realizations.

No learning experiment should proceed until these tests pass.

17.2 Stage 1: collect broad exploration trajectories

Use random/grid/heuristic actions to collect $\mathcal{D}_{\text{WM}}$ and a large pool of pilot I/Q windows. The World Models paper first collected random rollouts before separately training the VAE and recurrent model [1]; the same principle is used here.

17.3 Stage 2: JEPA pretraining

Train E_θ , P_ϕ , and EMA target encoder $E_{\bar{\theta}}$ on $\mathcal{D}_{\text{JEPA}}$ using (77). At the end, discard P_ϕ and retain the trained context encoder. Freeze JEPA for the first downstream experiments.

17.4 Stage 3: encode trajectories and train MDN-LSTM

For every transition, compute

$$\mathbf{z}_t = [E_\theta(\mathbf{x}_t^{\text{IQ}}); f_q(\mathbf{q}_t)]. \quad (103)$$

Train the MDN-LSTM using teacher forcing and (93). The World Models reference used 5 mixtures and 256 LSTM hidden units for CarRacing [1], which are the default values here.

17.5 Stage 4: generate exhaustive-search expert trajectories

Run the exhaustive-search oracle online in the simulator, producing coherent trajectories with labels $\mathbf{a}_t^*$. Encode each trajectory using the frozen JEPA and pass it sequentially through the frozen MDN-LSTM to obtain $\mathbf{h}_t$.

17.6 Stage 5: behavior-clone the controller

Train only controller parameters using (101). Evaluate first on held-out expert states and then in closed-loop simulator rollouts.

17.7 Stage 6: optional long-horizon fine-tuning

Fine-tune controller parameters in the real simulator using CMA-ES or PPO. Keep JEPA and the world model frozen for the cleanest first study. Later experiments may unfreeze only the JEPA projection layer or the final encoder stage.

17.8 Stage 7: optional dream-environment training

Once latent transition and reward prediction are accurate, sample

$$\hat{\mathbf{z}}_{t+1} \sim p(\mathbf{z}_{t+1} \mid \mathbf{z}_t, \mathbf{a}_t, \mathbf{h}_t) \quad (104)$$

and use the reward head $\hat{r}_t$ to train the controller in imagined trajectories. A temperature parameter may scale the mixture uncertainty, analogous to the stochastic dream-environment mechanism in [1]. This stage is explicitly optional and should not be attempted until real-environment performance and model calibration are established.

Algorithm 1 One Environment Step

Require: Environment state $\mathbf{s}_t^{\text{env}}$, action $\mathbf{a}_t = (P_t, P_j)$

- 1: Update/hold geometry and generate channel realization for slot t
 - 2: Compute $\gamma_H(t)$ and $\gamma_U(t)$ using (31)–(32)
 - 3: Compute $R_H(t)$, $R_U(t)$, $R_s(t)$ using (35)
 - 4: $g_t \leftarrow \mathbb{I}\{R_s(t) \geq R_s^{\min}\}$
 - 5: $\mu_H \leftarrow g_t \mathbb{I}\{R_H(t) \geq R_{\text{req}}\}$
 - 6: $\mu_U \leftarrow g_t \mathbb{I}\{R_U(t) \geq R_U^{\text{dec}}\}$
 - 7: Update $\Delta_H(t+1)$ using (40)
 - 8: Update $\Delta_U(t+1)$ using (41)
 - 9: Compute reward r_t using (56)
 - 10: Advance UAV mobility to $\mathbf{p}_U(t+1)$
 - 11: Generate next-slot pilot observation $\mathbf{x}_{t+1}^{\text{IQ}}$ using (20)
 - 12: Construct $\mathbf{q}_{t+1}$
 - 13: **return** $\mathbf{x}_{t+1}^{\text{IQ}}, \mathbf{q}_{t+1}, r_t$, diagnostics
-

18 Algorithms

19 Reference Dimensions

20 Reference Simulation Parameters

21 VAE Baseline for the Central Comparison

To test whether JEPA is responsible for any improvement, implement a VAE that receives exactly the same normalized I/Q observation. Let

$$q_\varphi(\mathbf{z}^{\text{RF}} \mid \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_\varphi(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_\varphi^2(\mathbf{x}))) \quad (105)$$

and decoder $p_\omega(\mathbf{x} \mid \mathbf{z}^{\text{RF}})$. Train with

$$\mathcal{L}_{\text{VAE}} = \mathbb{E}_{q_\varphi} [\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2] + \beta_{\text{KL}} D_{\text{KL}}(q_\varphi(\mathbf{z} \mid \mathbf{x}) \parallel \mathcal{N}(\mathbf{0}, \mathbf{I})). \quad (106)$$

For the fairest primary comparison, match

$$d_{\text{RF,VAE}} = d_{\text{RF,JEPA}} = 128, \quad (107)$$

and use the same side-information encoder, MDN-LSTM, controller, expert dataset, optimizer, and evaluation seeds. A secondary ablation may use the original World Models latent size $d_z = 32$ [1].

22 Evaluation Baselines and Ablations

22.1 Required baselines

At minimum, compare:

1. random power allocation;
2. fixed equal/hand-tuned power allocation;

Algorithm 2 Exhaustive-Search Expert Rollout

Require: Environment $\mathcal{E}$, power grid $\mathcal{A}_{\text{grid}}$

```
1: Reset  $\mathcal{E}$  and observe  $(\mathbf{x}_0^{\text{IQ}}, \mathbf{q}_0)$ 
2: for  $t = 0, \dots, T - 1$  do
3:    $J_{\text{max}} \leftarrow -\infty$ 
4:   for all  $\mathbf{a} \in \mathcal{A}_{\text{grid}}$  do
5:     Clone environment:  $\mathcal{E}' \leftarrow \text{Clone}(\mathcal{E})$ 
6:     Execute one counterfactual step in  $\mathcal{E}'$  with  $\mathbf{a}$ 
7:     Compute  $J_t(\mathbf{a}) = r_t(\mathbf{a})$ 
8:     if  $J_t(\mathbf{a}) > J_{\text{max}}$  then
9:        $J_{\text{max}} \leftarrow J_t(\mathbf{a}); \mathbf{a}_t^* \leftarrow \mathbf{a}$ 
10:    end if
11:  end for
12:  Execute  $\mathbf{a}_t^*$  in the real rollout environment  $\mathcal{E}$ 
13:  Store transition and expert label
14: end for
15: return  $\mathcal{D}_{\text{ES}}$ 
```

Algorithm 3 JEPA Pretraining

Require: Unlabeled I/Q dataset $\mathcal{D}_{\text{JEPA}}$

```
1: for each minibatch  $\mathbf{x}$  do
2:   Normalize and upsample  $2 \times 4 \times 256 \rightarrow 2 \times 256 \times 256$ 
3:   Sample latent mask  $\mathbf{M}_{\text{lat}}$ 
4:   Encode visible context  $\mathbf{H} = E_{\theta}(\mathbf{x}, \mathbf{M}_{\text{lat}})$ 
5:   Insert mask tokens and predict  $\hat{\mathbf{Y}} = P_{\phi}(\tilde{\mathbf{H}})$ 
6:   Compute dense target  $\mathbf{Y} = E_{\bar{\theta}}(\mathbf{x})$  without gradient
7:   Compute  $\mathcal{L}_{\text{JEPA}}$  on masked positions only
8:   Update  $\theta, \phi$  using AdamW
9:   Update  $\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau)\theta$ 
10: end for
```

3. one-step exhaustive-search oracle;
4. direct MLP controller from engineered state $\mathbf{q}_t$ only;
5. direct MLP controller from flattened/processed I/Q without a world model;
6. VAE $\rightarrow$ MDN-LSTM $\rightarrow$ same controller;
7. JEPA $\rightarrow$ controller without recurrent memory;
8. JEPA $\rightarrow$ MDN-LSTM $\rightarrow$ same controller (proposed);
9. proposed model after optional long-horizon controller fine-tuning.

22.2 JEPA-specific ablations

Evaluate:

- antenna vs. time vs. multi-block masking;

Algorithm 4 MDN-LSTM Training

Require: Ordered trajectories $\mathcal{D}_{\text{WM}}$, frozen JEPA E_θ

```
1: for each sequence batch do
2:   Encode all I/Q windows to  $\mathbf{z}_t^{\text{RF}}$ 
3:   Encode side information to  $\mathbf{z}_t^{\text{net}}$ 
4:   Concatenate  $\mathbf{z}_t = [\mathbf{z}_t^{\text{RF}}; \mathbf{z}_t^{\text{net}}]$ 
5:   Reset/initialize LSTM hidden and cell states at episode boundaries
6:   for  $t$  in sequence do
7:      $(\mathbf{h}_{t+1}, \mathbf{c}_{t+1}) \leftarrow \text{LSTM}([\mathbf{z}_t; \tilde{\mathbf{a}}_t], \mathbf{h}_t, \mathbf{c}_t)$ 
8:     Produce mixture parameters for  $p(\mathbf{z}_{t+1} \mid \mathbf{z}_t, \mathbf{a}_t, \mathbf{h}_t)$ 
9:     Accumulate negative log-likelihood
10:  end for
11:  Backpropagate through time and update  $\psi$ 
12: end for
```

Table 1: Proposed initial tensor and network dimensions. Values marked “proposed” are new implementation choices, not fixed by the source papers.

Quantity	Dimension	Comment
Raw I/Q tensor	$2 \times 4 \times 256$	WirelessJEPA-aligned input
Upsampled JEPA tensor	$2 \times 256 \times 256$	WirelessJEPA preprocessing
HAP receive antennas	$M_H = 4$	Required for multi-antenna I/Q
RF latent	$d_{\text{RF}} = 128$	Proposed; ablate 64/128/256
Side-information vector	$d_q = 8$	Proposed
Network-state embedding	$d_{\text{net}} = 32$	Proposed
Joint latent	$d_z = 160$	$128 + 32$
Normalized action	2	P_t, P_j
LSTM input	162	$d_z + 2$
LSTM hidden state	$d_h = 256$	World Models CarRacing-aligned
MDN mixtures	$K = 5$	World Models-aligned
Controller input	416	$d_z + d_h$
Controller output	2	transmit and jamming power

- native $2 \times 4 \times 256$ processing vs. $2 \times 256 \times 256$ upsampling;
- $d_{\text{RF}} \in \{64, 128, 256\}$;
- frozen JEPA vs. last-stage/projection fine-tuning;
- RF latent only vs. RF latent plus side information.

22.3 World-model ablations

Evaluate:

- no RNN (controller sees $\mathbf{z}_t$ only);
- deterministic LSTM next-latent regression;
- MDN-LSTM with $K \in \{1, 5, 10\}$;

- hidden size $d_h \in \{128, 256, 512\}$;
- different trajectory-data mixtures.

23 Performance Metrics

23.1 Wireless control metrics

Report at least

$$\bar{\Delta}_H = \frac{1}{T} \sum_t \Delta_H(t), \quad (108)$$

$$\bar{\Delta}_U = \frac{1}{T} \sum_t \Delta_U(t), \quad (109)$$

$$\bar{R}_s = \frac{1}{T} \sum_t R_s(t), \quad (110)$$

$$P_{\text{sec,out}} = \frac{1}{T} \sum_t \mathbb{I}\{R_s(t) < R_s^{\min}\}, \quad (111)$$

$$\bar{P}_t = \frac{1}{T} \sum_t P_t(t), \quad (112)$$

$$\bar{P}_j = \frac{1}{T} \sum_t P_j(t), \quad (113)$$

$$\eta_{\text{succ}} = \frac{1}{T} \sum_t \mu_H(t), \quad (114)$$

$$\eta_{\text{leak}} = \frac{1}{T} \sum_t \mu_U(t). \quad (115)$$

23.2 Representation metrics

To determine whether JEPA learns useful channel structure rather than merely improving final reward, use frozen-encoder probes for quantities available from the simulator, e.g.

- source AoA prediction;
- source-HAP large-scale gain prediction;
- instantaneous $\|\mathbf{h}_{SH}\|^2$ regression;
- SNR-bin classification;
- channel-condition or fading-class prediction.

These are diagnostic probes, not inputs required by the final controller.

23.3 World-model metrics

Report

- MDN negative log-likelihood on held-out trajectories;

- one-step latent prediction MSE using mixture mean;
- H -step open-loop latent rollout error;
- reward prediction MSE if the reward head is used;
- calibration/coverage of predicted latent uncertainty.

23.4 Control-learning metrics

Report

- BC power error: MSE/MAE relative to exhaustive-search labels;
- fraction of actions within 1 dB or 2 dB of the oracle;
- cumulative discounted reward;
- number of real-environment interactions required to reach a target performance;
- inference time per slot.

24 Recommended Experimental Progression

A disciplined progression will make failure analysis much easier:

1. **Physics only:** verify rate/secretcy/AoI curves over (P_t, P_j) grids.
2. **Exhaustive oracle:** confirm that the chosen objective produces sensible non-degenerate power allocations.
3. **JEPA only:** show that RF embeddings preserve channel/AoA information.
4. **World model only:** show predictable next-latent dynamics and meaningful uncertainty.
5. **BC controller without RNN:** establish a lower-complexity supervised baseline.
6. **BC controller with MDN-LSTM:** test whether memory improves expert imitation and closed-loop metrics.
7. **VAE vs. JEPA controlled comparison:** keep everything downstream identical.
8. **Long-horizon fine-tuning:** test whether RL/CMA-ES improves beyond the one-step expert.
9. **Mobility/OOD tests:** evaluate unseen UAV speeds, positions, fading severities, SNRs, and shadowing.
10. **Optional dream training:** only after model error and uncertainty are well characterized.

25 Implementation Sanity Checks

Before reporting results, enforce the following checks.

25.1 Unit checks

- All dBm quantities must be converted to linear watts before SINR calculation.
- Distances entering (10) must be in km.
- Bandwidth entering the noise calculation must be in Hz.
- Rate thresholds must use the same units as R_H , R_U , and R_s .

25.2 Causality checks

At action time t , the controller may use only information available before the data action:

$$\mathcal{O}_t = \{\mathbf{x}_t^{\text{IQ}}, \mathbf{q}_t, \mathbf{h}_t\}. \quad (116)$$

It must not use $R_H(t)$, $R_U(t)$, $R_s(t)$, or post-action I/Q samples to select $\mathbf{a}_t$.

25.3 No privileged-state leakage

Exact instantaneous $h_{SU}(t)$ and $h_{HU}(t)$ should remain privileged simulator variables unless the experimental assumption explicitly grants them to the agent. If side information contains β_{SU} and β_{HU} , these should be interpreted as tracked/estimated large-scale gains, not exact instantaneous small-scale coefficients.

25.4 Data leakage checks

Do not place windows from the same contiguous trajectory in both training and test sets. For OOD testing, hold out entire UAV trajectories, channel seeds, SNR ranges, or fading-parameter ranges.

26 Open Design Decisions to Revisit After the First Smoke Runs

The framework above is fully implementable, but the following items should be treated as experimental design variables rather than fixed truths:

1. exact pilot waveform and pilot/data duration split;
2. whether the $2 \times 4 \times 256 \rightarrow 2 \times 256 \times 256$ repetition used by WirelessJEPA is beneficial for control;
3. JEPA compact latent size d_{RF} ;
4. whether exact AoLI is observable or must be estimated;
5. one-step versus short-horizon exhaustive-search labels;
6. hard secrecy gate versus soft secrecy penalty;
7. controller choice: linear, one-hidden-layer MLP, CMA-ES, or PPO;
8. whether MDN prediction should cover the entire joint latent or only the RF latent with separate explicit scalar-state heads;
9. temporal channel correlation model needed to make history genuinely informative;
10. whether final JEPA layers should be fine-tuned after the frozen-encoder baseline is established.

27 Minimum Reproducible Experiment

A first reproducible end-to-end experiment can be defined as follows:

1. Static HAP at 20 km and static UAV at 5 km horizontal source distance.
2. Four-antenna HAP ULA, $f_c = 28$ GHz, $B = 20$ MHz, Nakagami fading.
3. Fixed pilot power $P_p = 10$ dBm and 256 pilot samples per slot.
4. $P_t, P_j \in [0, 30]$ dBm; exhaustive grid at 1 dB increments.
5. Generate 100k–1M unlabeled I/Q windows across randomized fading, AoA, shadowing, and SNR for JEPA pretraining.
6. Pretrain WirelessJEPA-style encoder for 100 epochs with multi-block masking; retain a 128-dimensional RF latent.
7. Generate 5k–10k trajectories of length 100–500 using mixed random/grid actions for the MDN-LSTM.
8. Train a 256-hidden-unit, 5-mixture MDN-LSTM.
9. Generate expert trajectories using one-step exhaustive search.
10. Train a linear two-output controller by BC.
11. Compare against a dimension-matched VAE using identical world model, controller, datasets, and test seeds.
12. Report AoI, AoLI, secrecy rate, secrecy outage, power consumption, world-model NLL, and controller oracle gap.

The exact dataset sizes above are engineering starting points, not source-paper requirements; they can be reduced for smoke tests and scaled after the pipeline is stable.

28 Summary of the Proposed Scientific Contribution

The proposed study is not simply “replace VAE by JEPA.” Its complete scientific question is:

Can a self-supervised, multi-antenna latent-prediction representation produce a more predictive and control-sufficient internal wireless state than a reconstruction-based VAE, such that an action-conditioned world model and compact controller can achieve better freshness–secrecy power allocation under partial RF observations?

The controlled experiment is therefore

VAE JEPA $\longrightarrow$ $\longrightarrow$ same MDN-LSTM same MDN-LSTM $\longrightarrow$ $\longrightarrow$ same controller same controller	(117)
--	-------

with identical wireless trajectories, expert labels, action space, reward, test seeds, and downstream optimization procedure. This design allows improvements to be attributed much more convincingly to the learned wireless representation rather than to unrelated changes in the controller or simulator.

References

- [1] D. Ha and J. Schmidhuber, “World Models,” *arXiv preprint arXiv:1803.10122*, 2018.
- [2] V. Chu, O. Mashaal, and H. Abou-Zeid, “WirelessJEPA: A Multi-Antenna Foundation Model using Spatio-temporal Wireless Latent Predictions,” *arXiv preprint arXiv:2601.20190*, 2026.
- [3] M. Sadiq, M. S. Haider, A. Fatima, M. S. J. Solaija, H. Jung, and S. A. Hassan, “Intent-Driven Hierarchical DRL for Secrecy-Aware AoI–AoLI Optimization in RIS-Assisted HAP-IoT Communications,” *IEEE Internet of Things Journal*, vol. 13, no. 9, pp. 18351–18362, May 2026, doi: 10.1109/JIOT.2025.3648959.

Table 2: Initial simulation parameters. “Source-aligned” indicates that the scale/value is inherited from [3]; “proposed” indicates a new baseline choice that should be swept.

Parameter	Initial value	Provenance / note
Carrier frequency f_c	28 GHz	Source-aligned
Bandwidth B	20 MHz	Source-aligned
Slot duration T_s	1 s	Source-aligned
HAP altitude H_H	20 km	Source-aligned
Initial source-UAV horizontal range	5 km	Source-aligned scale
Atmospheric attenuation α_{atm}	0.06 dB/km	Source-aligned
UAV extra shadow margin	6 dB	Source-aligned
Noise density N_0	−174 dBm/Hz	Source-aligned
Noise figure NF	5 dB	Source-aligned
Source antenna gain	8 dBi	Source-aligned
HAP effective receive gain	20 dBi	Source-aligned; avoid double-counting array gain
UAV receive gain	1 dBi	Source-aligned
Residual SI factor κ	0.35	Source-aligned
Source-HAP Nakagami m_{SH}	3	Source-aligned
Source-UAV Nakagami m_{SU}	2	Source-aligned
HAP-UAV Nakagami m_{HU}	2	Proposed starting value
Transmit power P_t	0–30 dBm	Source-aligned range
Jamming power P_j	0–30 dBm	Source-aligned range
Pilot power P_p	10 dBm	Proposed; fixed before decision
Pilot length W	256 samples	WirelessJEPA-aligned
Packet size L_{pkt}	2 Mbit	Source-aligned frame size
AoI/AoLI cap Δ_{max}	200 slots	Proposed numerical cap
Secrecy threshold R_s^{min}	0.5 Mbit/s	Proposed starting value; sweep
UAV decode threshold R_U^{dec}	R_{req}	Proposed
Power-grid spacing	1 dB	Proposed; 961 actions/state
JEPA epochs	100	WirelessJEPA-aligned
JEPA optimizer	AdamW + cosine decay	WirelessJEPA-aligned
EMA momentum τ	0.996 → 1.0	WirelessJEPA-aligned
LSTM hidden units	256	World Models-aligned
Gaussian mixtures K	5	World Models-aligned
Discount factor γ	0.99	Proposed / common RL starting value